

Center for Multiscale Modeling of Multiphase Combustion

Predictive Simulation Centers (PSCs)

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1 Project Objectives

The University of Florida (UF), in collaboration with the Georgia Institute of Technology (GT) and Purdue University (PU), proposes to establish a Center for Multiscale Modeling of Multiphase Combustion (CM³C) as part of the NNSA’s Predictive Science Academic Alliance Program-IV’s broad single science/engineering-focused Predictive Simulation Center (PSC). We propose solving the problem of *phase change and group combustion* of a dense cloud of particles under high-speed and detonation conditions at an unprecedented level of physical detail with a state-of-the-art multiscale modeling framework that combines the powers of scientific computing (SC) and machine learning (ML). The key technical objective of the center is to predict the spatio-temporal higher-order statistics of phase change and combustion in polydisperse multiphase systems in high-speed environments through exascale computation. The intellectual objectives of the proposed work are to (i) understand group combustion via a multiscale approach, where the micro-meso-macroscales are linked by hierarchical physics-informed ML models that will enable truly predictive large-scale simulations; (ii) develop a trace-based load-balancing strategy for efficient coupling of the gas and particulate phases using the exascale codebase MFC (Multicomponent Flow Code); (iii) develop a multiscale modeling framework where predictive simulations of the demonstration problem will be validated with uncertainty budget (UB)-driven model improvements achieved through accompanying micro/mesoscale experiments/simulations; (iv) develop workflow strategies for seamless exchange of information and data between experiments, simulations, ML models, and uncertainty quantification (UQ) across scales; and (v) educate students and staff in the science of exascale simulation and UQ and place them at the NNSA Labs.

2 Scientific and Technical Merit

2.1 Overarching Vision and Integrated Problem

The group combustion of a cloud of particles in a high-speed environment is a formidable problem, as it combines turbulence, multiphase flow, shock waves, and reactive flow physics. The wide range of scales, phase changes, and stiff chemical reactions make direct numerical simulations (DNS) of combustion of even an isolated particle in a high-speed environment a problem worthy of exascale computation. The collective burning of a cloud of particles cannot be approached as a superposition of individual burning processes. Particle-gas interactions at the mesoscale through preferential fingering, particle-gas-particle interactions at the microscale in the form of localized oxidizer depletion in reacting wakes, and close-range particle-particle interactions through collisions render the problem far more complex, and multiscale methods are required for efficient use of exascale facilities.

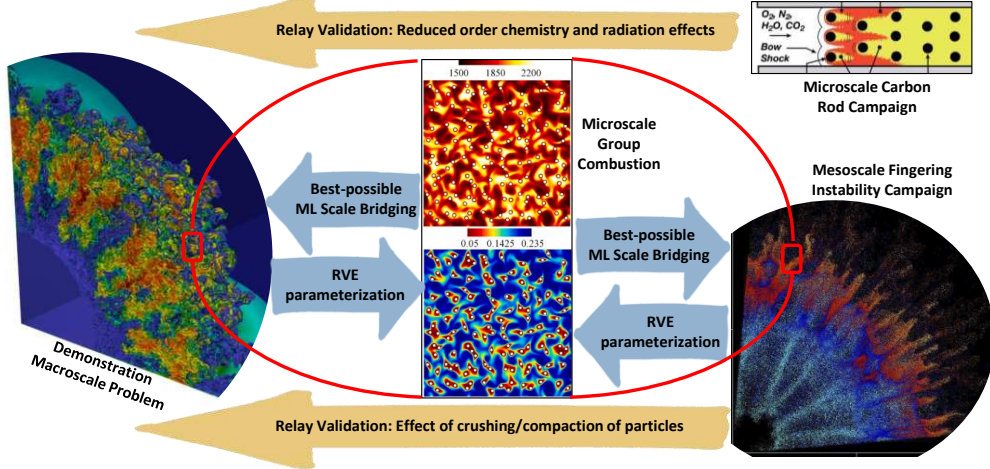


Fig. 1: Schematic of the proposed approach. Euler-Lagrange (EL) and Euler-Euler (EE) simulations of the demonstration problem of explosive dispersal of burning particles will employ *best-possible* machine learning (ML)-based closure models developed using particle-resolved (PR) microscale-informed understanding of *group combustion* [1] within all possible representative volume elements (RVE). Rigorous uncertainty quantification will be achieved through *relay validation* of reduced-order chemistry and other hard-to-model complex physics using additional micro and mesoscale campaigns [2].

We propose to develop a bold new hierarchical scale-bridging approach for high-fidelity closure models by providing proper context to the bottom-up approach currently employed in complex multiscale multiphase problems (see [fig. 1](#)). Instead of building closure models based only on the canonical configuration of an isolated burning particle, we inform the coarse-grained models with a complete statistical description of all possible group combustion configurations. To achieve this goal, we leverage advancements in AI, melding SC with ML at multiple scales. We advance a co-design strategy that combines particle-resolved (PR), Euler-Lagrange (EL), and Euler-Euler (EE) simulations of high-speed combustion of dense particle clouds with physics-informed generative and operator-learning models. The following attributes distinguish CM³C:

1. **Exascale-enabled Discovery of Complex Physics:** The complex nature of group combustion under high-speed and detonation conditions makes controlled experiments with diagnostics challenging. We hypothesize that fully validated coarse-grained exascale simulations are the best pathway for fundamental breakthroughs that can lead us out of current empiricism.
2. **Scale-bridging with physics-informed ML:** We propose a novel Lagrangian-based multiscale modeling approach that aims to rigorously develop the *best-possible* closure models of group combustion. We argue that this ambitious scale-bridging goal can be achieved only by melding ML and SC in two steps: (i) generative neural network models are used to rapidly generate vast amounts of surrogate data that rival high-fidelity exascale simulations in accuracy, and (ii) operator-learning networks are then used for learning large-parametric closures.
3. **UB-driven simulation roadmap:** We will employ innovative *relay validation* for quantification of uncertainties in the macroscale EL simulations of the demonstration problem through companion micro/mesoscale simulation/experimental campaigns (see [fig. 1](#)). We will use uncertainty budgets (UB) to rank order the sources of uncertainties and inform the next cycle of simulations and experiments for uncertainty reduction (UR) and model improvement using *active learning*. The mesoscale campaigns also serve to test the broader applicability of the closure models in diverse configurations.
4. **Dynamic load balancing for exascale performance:** For PR, EL, and EE simulations,

we will use the exascale codebase MFC (Multicomponent Flow Code [3, 4]). We advance the development of the bin-based load balancing strategy and the trace-based workload prediction tool for exascale EL simulations. The code is ready for performant simulation on the MI300A nodes of El Capitan and Tuolumne.

Figure 2 shows the schematic of our demonstration problem consisting of a high explosive (HE) hemispherical charge surrounded by reactive carbon (C) or magnesium (Mg) particles. Detonation of the HE rapidly shocks, heats, and accelerates the particles. As the particulate front radially expands, it undergoes complex instabilities and reactions [5–8]. Molten Mg burns within vapor diffusion flames [9–11] while carbon undergoes heterogeneous surface and gas-phase reactions [12–14]. Thus, C and Mg are selected to capture these two ends of particle burning regimes. Predicting the reactive particle-gas and particle-gas-particle interactions across the broad range of scales within the expanding post-detonation flow poses an immense computational challenge. Meeting this challenge is the central goal of the proposed work.

The proposed center will prioritize integrated simulations of the demonstration problem from the outset using MFC (Multicomponent Flow Code [3, 4]), which has demonstrated near-roofline performance on OLCF Summit (V100), OLCF Frontier (MI250X), and LLNL El Capitan early access system Tioga (MI300A). The hero simulations of the demonstration problem will be performed annually with increased computational resources and improved modeling fidelity (to be later discussed in table 2). Relay validation and propagation of all the contributing modeling errors and uncertainties will rigorously quantify the overall uncertainty in the prediction metrics. We will assess the UB annually to identify the modeling and numerical weaknesses that most contribute to uncertainty. This strategy enables uncertainty reduction through improvements to ML models and numerical methods.

2.2 Predictive Science Plan

Combustion and phase change processes for droplets and particles are important to many applications, including energy, propulsion, national security, and industrial safety. Many scenarios occur at short time scales under extreme and optically thick conditions. Understanding the explosion hazards of reactive particulate clouds or improving the performance of propulsion and explosive systems requires numerical simulations alongside experiments. Predictive simulations of multiphase reactive flows must represent the chemical energy release processes of burning particles. This precision requires tackling the multi-faceted challenges of compressible flow features, thermal transients, reacting wakes, classical/pseudo turbulence, random distribution of particulate clusters, and heterogeneous/gas-phase reactions at a broad range of time scales. The underlying multiphase reactive flow physics is important in many other applications of interest to the NNSA, including multiphase explosives [15–18], ejecta dynamics [19–22], environmental influence in hypersonic flight [23, 24], etc., [25, 26].

2.2.1 Multiphase Group Combustion Physics

In the demonstration problem (fig. 2), we will consider a HE whose detonation properties are well-understood so that we can focus on predicting post-detonation effects. In the fig. 2 configuration, an intense shock wave followed by a contact interface between the detonation products and the air first propagates through the surrounding particles. The shock exits the particles as an intense air blast

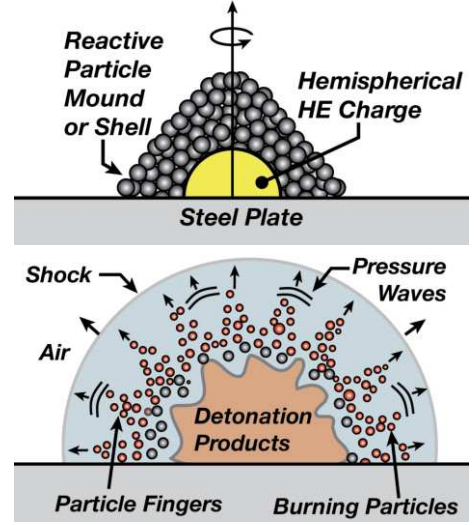


Fig. 2: Schematic of the demonstration problem.

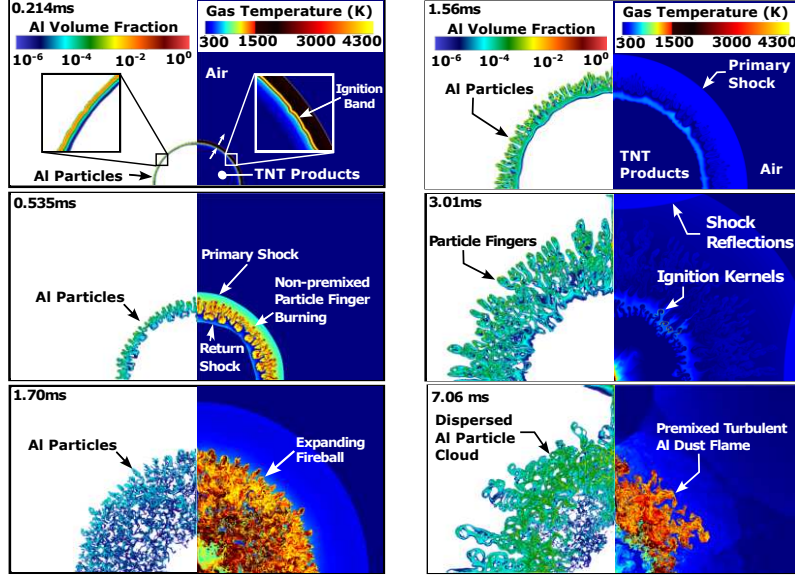


Fig. 3: Ignition and combustion of a TNT-dispersed Al particle shell for two shell thicknesses [8].

while the particles accelerate outward and disperse radially. Complex time-dependent shock-heating and mixing leads to delayed ignition and combustion of the dispersed particles [5, 7].

Initially, the particles travel slower, and the high-speed relative velocity creates a complex system of interacting shocklets. Each particle is surrounded by a high temperature fuel-rich flame envelope. The burn rate of a particle is substantially influenced when caught within the flame envelope of another [27]. Under dense conditions, the flame envelopes merge into a larger collective flame envelope, whose behavior is not a superposition of individual flame envelopes. A meaningful at-a-distance interaction exists between the burning particles, termed *group combustion*. Since these interactions will not be resolved at the macroscale, their effects must be accurately captured in the closure models. Empirical approaches such as group combustion number have been proposed [28]. However, an approach that deterministically accounts for inter-particle interactions is needed.

As particles are propelled outward, they result in various structures (see fig. 3). The two aspects of instabilities are the initiation (or receptivity) mechanisms that dictate the onset and the linear/nonlinear growth mechanisms [29]. The propagation of intense force chains has been studied as the primary initiation mechanism. The subsequent growth is due to Rayleigh–Taylor (RT), Richtmyer–Meshkov (RM), and other non-classical instabilities. Due to delayed ignition, particle reactivity does not influence the early onset of perturbations. However, later time heat release is expected to influence the RT and RM instabilities. The instabilities alter the spatio-temporal distribution of burning particles and impact quantities of ultimate interest, such as turbulent mixing, reaction rate, and front velocity.

Figure 3 shows our coarse-grained EE results of explosively-dispersed Al particles, which display a wide variety of ignition and combustion phenomena [8]. When the particle-to-charge mass ratio is low, the particles ignite promptly in the shock-heated air and burn as non-premixed flames on the outer edge of the expanding fingering instabilities of the particle cloud (left frames). When the mass ratio is large, the shock-heated air temperature is too low to ignite the particles. Rather, the particles undergo delayed ignition at the inside edge of the cloud and burn as a turbulent premixed dust-cloud during the negative phase of the blast (right frames). Despite these simulations, existing subgrid models that are based on single-particle combustion cannot capture the group

effects accurately. For predictive capability we require next-generation microscale-informed closure models of unprecedented fidelity.

2.2.2 Science Questions and Prediction Goals

The key technical objectives are to accurately predict (i) the instability-driven spatial clustering of the explosively dispersing particles, (ii) the time-dependent burn-rate distribution of the dispersing particles, and (iii) the multiphase afterburn effect on the blast wave dynamics. We desire to achieve these objectives in the broadest possible manner with a universally applicable physics-based modeling framework whose predictive capability applies to a wide range of applications. To achieve these objectives, we set out to answer the following science questions:

1. In group combustion, the average burn rate of a cluster of particles is expected to be lower due to oxidizer depletion within the combined flame envelope. The current understanding of group combustion is limited to laminar near-quiescent environments [30]. As in the force modeling of particle-gas-particle interactions [31, 32], group combustion becomes complicated with added convective [1, 27, 33–36] and compressible effects. The physics of group combustion in the presence of nonlinear interactions through mass, momentum, energy exchanges, and chemical reactions remains to be explored.
2. The unsteady effects of group combustion, such as shock-induced ignition and burning of a cloud of particles, await investigation. The unique physics of shock-induced force on a particle cluster has been revealed in recent years [37–39]; the proposed work will help understand the corresponding particle-to-particle variations in heating, ignition, and burn-rate as shocks propagate through clouds of reactive particles. Understanding microscale physics is central to their accurate modeling in macroscale simulations.
3. Multiphase explosions are complicated due to the many interacting mesoscale effects: instability-induced clustering and mixing/burning of the clusters [8, 40–42]. The current understanding of how these mesoscale inhomogeneities conspire to produce macroscale effects is only qualitative. Understanding the effects of mesoscale physics of clustering on group combustion is needed.
4. The classical single-phase RT and RM instabilities have been well studied [29]. The proposed work will strengthen our understanding of the early onset of perturbations within particulate beds and the subsequent growth of multiphase RT and RM instabilities. The proposed work will explain how group combustion influences RT and RM instabilities and contributes to other instability mechanisms unique to multiphase systems.

Given the formidable nature of the demonstration problem and the associated challenges of validation-quality measurements, we layer the prediction goals into basic and higher-order metrics. The basic quantities of interest (QoI) for validation are (Q1) the blast wave location as a function of time, (Q2) time-dependent pressure at selected radial distances over repeated experiments (realizations), and (Q3) the volume fraction of reactive particulate at selected radial locations. Higher-order metrics include (Q4) temperature and emission from combustion at selected radial distances, (Q5) the number and amplitude of fingering instabilities in the expanding cloud of burning particles, and (Q6) time-dependent PDFs of pressure and particulate volume fraction at selected radial locations.

2.2.3 Exascale Simulations of the Demonstration Problem

The wide range of length and time scales associated with the demonstration problem (referred to as the *macroscale*) shown in [fig. 2](#) is self-evident. At this scale, a fully-resolved approach that predicts the flow around each micron-sized particle is impossible. We must instead turn to the Euler-Lagrange (EL) approach, where the gas phase governing equations are solved on an Eulerian mesh while the particles are tracked in the Lagrangian frame. In the proposed configuration (see [fig. 2](#)), the hemispherical charge will be surrounded by a conical mound of $O(10^{11})$ to $O(10^{13})$ particles. The

corresponding exascale EL-simulations will consider a computational domain of size $1 \times 1 \times 1 \text{ m}^3$, which is chosen to capture the blast wave and the instabilities. An exascale discretization with $O(10^{12})$ grid points yields an effective resolution of $100 \mu\text{m}$ (the actual grid will be non-uniform to capture the shock and the particle front).

The initial mound of particles will be sieved to be a narrow spectrum of particle size, which will be replicated in the EL simulations. As they accelerate outward, the reaction rate will vary from particle to particle, and the size spectrum is anticipated to broaden. The particles are subgrid, with the largest particle much smaller than the EL grid. The non-uniform grid will be chosen so that all the EL grid cells contain at most $O(100)$ particles. We have already performed and experimentally validated simulations of the detonation of the bare hemispherical charge and the turbulent mixing of the post-detonation flow [43–46]. However, progressing to predictive simulations of the corresponding reactive multiphase flow poses a grand challenge. Thus, the central focus of the proposed work is to enable macroscale simulations whose results can be trusted to provide reliable answers to the science questions presented in §2.2.2. Towards this goal, we start by recognizing that in the EL simulations, the underlying gas-phase governing equations are spatially filtered (or averaged) on a scale that is an order of magnitude larger than the size of the particles. The penalty of spatial averaging is threefold: (i) the momentum/heat/mass transfer and surface chemistry must be modeled in terms of force/Nusselt/Sherwood number and surface reaction rate models. (ii) The effect of unresolved dynamics, which include shocklets and flame envelopes, on the resolved scales must be modeled as Reynolds stresses, Reynolds fluxes, and sink/source of gas-phase reactants/products. (iii) We must represent the surface and gas-phase chemistry with a limited set of reduced-order reactions.

2.2.4 Best-possible Closures from Microscale Simulations

Our multiscale modeling strategy is built upon ML to arrive at the best-possible EL simulations of the demonstration problem. This strategy relies on the following five assertions:

1. The filtered EL governing equations do *not* pose a problem since they can be derived rigorously, without approximations [47]. The real challenge is the subgrid closure models (of force, heat/mass transfer, Reynolds stress, flux, etc.) and the reduced-order surface/gas-phase reactions.
2. Subgrid turbulence can be separated into pseudo turbulence generated by the reactive particles and classical cascading turbulence. At modest particle Reynolds numbers ($\text{Re} \lesssim O(10^2)$), subgrid pseudo turbulence can be deterministically modeled in terms of computed EL quantities.
3. Pseudo turbulence dominates at the early stages of explosive dispersal. At later stages of fingering instability, classical turbulent mixing becomes relevant. Unlike pseudo turbulence, classical turbulence is stochastic and cannot be uniquely inferred from the computed EL quantities.
4. There exists *best-possible* closure models, which, when used, will lead to the best-possible EL simulations, whose time evolution will statistically replicate the true dynamics. The best-possible EL simulation must accurately infer the deterministic effect of pseudo turbulence while statistically inferring the effect of subgrid classical turbulence.
5. Reduced-order chemical reactions can be developed and validated in the context of a small collection of particles undergoing group combustion.

Fully resolved microscale simulations focusing on the details of group burning of a cluster of $O(10^3)$ particles in a high-speed cross-flow will be pursued. Here, we refer to these as *high-fidelity particle-resolved* (HF-PR) simulations. Based on the expected range of particle Reynolds and Mach numbers, a conservative estimate will require $O(10^2)$ grid points per particle diameter [1, 32], yielding an overall grid requirement of $O(10^{11})$. These HF-PR simulations will also serve as the test bed for developing reduced-order surface and gas-phase chemical reactions for C and Mg particles. Despite adequate resolution, the HF-PR simulations will involve uncertainty due to reduced-order chemistry

and other approximations to be elaborated in §2.3.5.

The best-possible closure models require a complete understanding of the microscale physics (as obtained in the HF-PR simulations) for all possible group combustion configurations encountered in the demonstration problem. However, the proposed HF-PR simulations will only be performed for a few group combustion configurations that can be afforded. *The key innovation of the proposed work is to use ML tools to bridge this gap in two steps.* In the first step, the HF-PR results will be used to train generative ML models that approach HF-PR simulations in accuracy. The cost-effective ML models will generate many HF-PR-like surrogates, providing complete coverage of all relevant group combustion configurations. The HF-PR and surrogate results will then be used in the second step to train operator learning ML models that serve as the best-possible closures in the EL simulations (to be discussed in §2.5).

2.3 Verification & Validation/Uncertainty Quantification Plan

Thorough validation and uncertainty quantification (UQ) of *all aspects* of explosive dispersal of reactive particles is a formidable challenge that is arguably beyond current reach. We will focus on validating the basic and higher-order QoIs in the demonstration problem presented in §2.2.2. Macroscale demonstration experiments and validation-quality measurements are described in §2.3.1. To quantify the uncertainty of the QoIs, all the contributing errors and uncertainties in the microscale simulations and ML closure models will be evaluated and propagated via scale-bridging. This process is described in §2.3.3 to §2.3.5.

Three complex physical processes require attention: (i) surface and gas-phase chemical reactions, (ii) particle and gas-phase radiation, and (iii) fragmentation/compaction of particles due to the blast wave. The proposed work will emphasize the modeling and UQ of chemical reactions of C and Mg particles. Although radiation and fragmentation/compaction are important in many applications, we will purposely consider configurations where radiation and fragmentation/compaction are less important such that their contribution to UQ of QoI is minimized.¹ Our choice of these macroscale configurations will be guided by our past experience [41, 42].

We propose employing the novel concept of *relay validation* where three micro/mesoscale campaigns are used to support UQ of reduced-order surface and gas-phase chemistry models. These experiments will also allow us to confirm our hypotheses on the criticality of radiation and fragmentation/compaction (see fig. 1). The micro/mesoscale experiments and relay validation are described in §2.3.2. The UQ of the companion micro/mesoscale simulations is presented in §2.3.5. Our plan for mitigating the risk of experiments failing to meet validation requirements is addressed in ??.

2.3.1 Macroscale Demonstration Experiments

Validation measurements of the demonstration problem will be conducted at Purdue University (PU) and will consider scaled versions of proven multiphase blast configurations (fig. 2) [5, 7, 48, 49]. The charge will consist of a 3 cm diameter hemisphere of pentaerythritol tetranitrate (PETN) surrounded by C and Mg particles. PETN is chosen for two reasons. First, PETN is nearly stoichiometrically balanced and produces little soot [50]. We thus mitigate the challenge of differentiating added particles from soot [48]. Second, the bare PETN charge configuration has been extensively investigated as part of a Sandia Grand Challenge LDRD thrust whose results are exemplified in fig. 4. The top row shows select temperature contours from reactive large eddy simulations that incorporate gas-phase kinetics with 59 species and 368 reactions [43]. These predictions were validated against experimental pressure [44], schlieren [46], planar laser-induced fluorescence (PLIF) [46], laser absorption spectroscopy

¹For instance, only the innermost layer of particles undergo fragmentation/compaction. These particles tend to remain entrained in the detonation products and react minimally. Therefore, with a sufficiently large mound of particles, the impact of fragmentation/compaction on QoI can be made small.

(LAS) [44], and bomb calorimetry [43] measurements. Favorable comparisons between simulations and measurements of pressure, temperature, and H_2O mole fraction are presented in [fig. 4](#). These data were acquired in a specially constructed blast-hardened LAS probe [44, 45, 51] which will be leveraged in the proposed work.

To ensure that the initial particle configurations can be accurately quantified and measured, the particles will be assembled in a free-standing mound. Before each shot, simple and accurate multiview image registration techniques will quantify the as-built profiles. Measures of the post-detonation QoI will be acquired using our suite of diagnostics, which have been proven viable for these harsh environments. Shock structure and strength will be quantified by schlieren imaging and pressure probes [43, 46, 50]. Spatially resolved particle concentrations will be quantified by tomographic inversion of visible [50] and X-ray extinction [7]. Finally, temperature and species concentrations (CO , CO_2 , H_2O) will be quantified by LAS within blast-hardened probe configurations [44, 45, 51]. UQ will use multiple measurements and repeated experiments.

Demonstration experiments at PU will be conducted in at least two campaigns. Tests in the first year will quantify blast pressures and flow structures using well-established diagnostics. These results will provide validation data for QoI (Q1) to (Q3) listed in [§2.2.2](#). Subsequent advancement of simulations will inform detailed placement of blast-hardened LAS gauges for higher-order QoI (Q4 to Q6), which will be conducted at roughly the project midpoint. In addition to the PSAAP supported experiments at Purdue, we anticipate opportunities to partner with the Naval Surface Warfare Center, Crane Division and/or the Air Force Research Laboratory, Munitions Directorate, to perform additional and larger-scale explosive experiments. If realized, the results will provide QoI that are above and beyond those discussed here (see ??).

2.3.2 Microscale and Mesoscale Experiments for Relay Validation

Microscale experiments will investigate the combustion of several closely-spaced, mm-sized carbon rods within a high-speed airflow ([fig. 5 \(a\)](#)). Flow velocities up to Mach 1.5 and stagnation temperatures of 500 K will mimic some of the most extreme relative air conditions experienced by the fastest moving particles in the explosive configuration. Individual carbon rods will be placed within this flow and Joule-heated to control surface temperatures. Videography will quantify the time-resolved changes in the surface profiles and overall reaction rates. This configuration is inspired by experimental methodologies, which are well-established for low-speed, isolated particle configurations [52–55] where results have already been proven useful for validation of HR-PR simulations [56–60].

Our relay validation will proceed as follows. In year one, we will study an isolated carbon rod at low flow rates to validate results against published literature [52–55]. These, and experiments extended

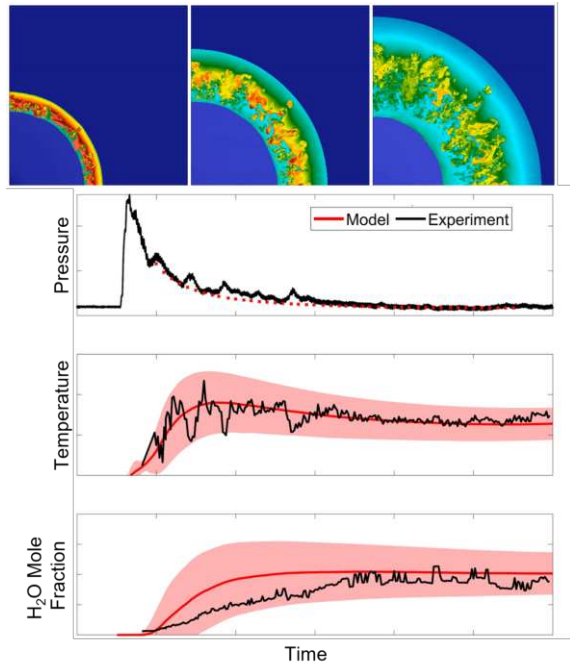


Fig. 4: Post-detonation results from bare PETN hemispheres: (top) select temperature contours from LES [43] (bottom) validation of pressure, temperature, and H_2O mole fraction at a fixed radial distance in the fireball [44, 51].

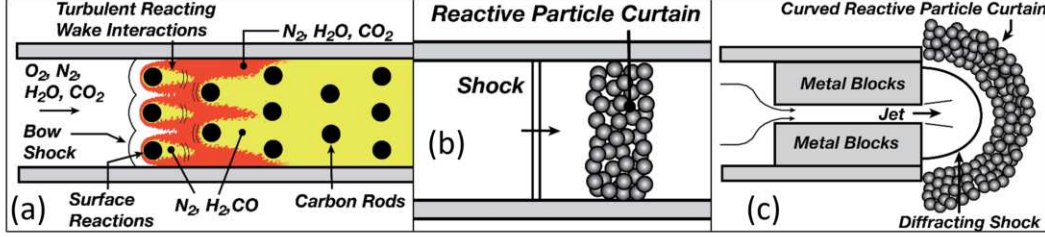


Fig. 5: Schematic of the (a) microscale, (b) planar mesoscale, and (c) cylindrical mesoscale configurations. Simulations will employ particle-resolved (PR) methodology at the microscale and Euler–Lagrange (EL) methodology at the mesoscale.

to higher flow velocities, will allow validation of companion HF-PR simulations of the rod geometry. Further experiments with a structured array of a few rods will allow rigorous quantification of reduced versus detailed surface and gas-phase chemistry uncertainties. Following this, we will expand capabilities and quantify novel multi-rod configurations whose positions and spacing are guided by UQ. These microscale experiments will leverage existing flow facilities [61] and Purdue’s world leading expertise in optical diagnostics to quantify higher order QoI using techniques such as CO, CO₂, and H₂O LAS [44, 45, 62], particle image velocimetry [63, 64] and PLIF [46].

Intermediate mesoscale investigations that bridge the micro and demonstration scales are crucial to relay validation. With the mesoscale configurations, we investigate the effect of group combustion’s pseudo-turbulence on multiphase instabilities and the self-organization of particles into clusters and voids. More importantly, by avoiding fragmentation/compaction, the mesoscale campaigns will allow estimation of the importance of these added mechanisms in the demonstration problem. Two mesoscale configurations will be considered (fig. 5(b,c)). (i) The ignition and group combustion of a gravity-fed planar curtain of bi-disperse particles subject to intense planar shock (fig. 5(b)) is being experimentally investigated at Sandia’s multiphase shock-tube [65–69]. These experiments use shocks with post-shock temperatures over 2500 K to disperse and ignite the reactive Al particles. High-speed camera imaging detects the particle curtain’s upstream and downstream edges as a function of time. In addition, cameras with bandpass filters detect AlO emissions to determine when, where, and if the Al particles are burning. (ii) A novel gravity-fed cylindrical particle curtain would further enhance these experiments as shown schematically in fig. 5(c). In this radially expanding configuration, fingering instabilities will be triggered, which would again be quantified with the suite of diagnostics for precise investigation of mesoscale mixing and reaction. These mesoscale experiments will use Mg and C particles and benefit from the broad diagnostic expertise at Sandia [70–74]. These configurations also validate the broader applicability of the ML-based closure models beyond the demonstration problem.

2.3.3 Scale-Separated Exascale Simulations

Exascale simulations will be performed at all three scales using MFC. HF-PR simulations will be performed at the microscale by solving the complete gas and particulate phase governing equations (block 1A of fig. 6). The heterogeneous surface and gas-phase reactions will be considered with the optimal reduced-order reactions. As in the experiments, the first set of simulations will consider stationary carbon rods. With the uncertainties of reduced-order chemistry established, microscale simulations will be performed for the reactive C and Mg particles (block 1B of fig. 6). Mesoscale EL simulations of the two configurations presented in fig. 5(b,c) will solve the filtered governing equations (block 2A). The subgrid closure models will initially be the best currently available. From year 3 onward, the EL simulations will employ the new ML-based closure models (block 1E). We will perform mesoscale simulations (block 2B) and validate against experimental results.

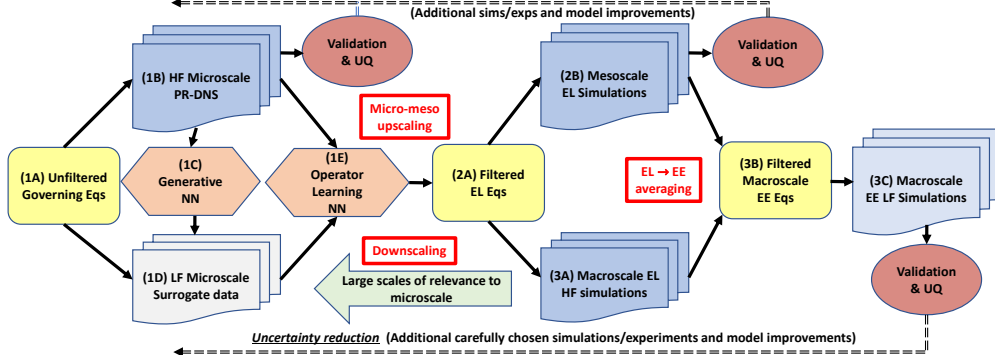


Fig. 6: Integration of operator-learning and generative neural networks with scale-separated exascale simulations. HF/LF: high/low-fidelity, UQ: uncertainty quantification, EL: Euler-Lagrange, EE: Euler-Euler, NN: neural network.

We will perform two types of macroscale simulations. The largest possible hero simulations of the demonstration problem will be performed yearly with increased computational resources and improved modeling fidelity (see Simulation Roadmap in §2.6.1). These EL simulations (black 3A) will employ the ML-based best-possible closure models *relayed* from the mesoscale. We will also conduct many low-fidelity EE simulations (block 3C) by solving the filtered gas and particulate phase governing equations (block 3B) [47]. Ensemble averaging the ML-based EL closure models (block 1E) over appropriate particle distributions will obtain the best possible EE closure models.

2.3.4 Mathematical Formulation of Scale Bridging

The governing equations, $\mathcal{F}$, and boundary conditions, $\mathcal{B}$, of the HF-PR are expressed as:

$$\mathcal{F}(\mathbf{u}(\mathbf{x}, t), \mathbf{v}(j, t), \lambda) = 0 \quad \text{and} \quad \mathcal{B}(\mathbf{u}(\mathbf{x}, \lambda) = 0 \quad \text{for} \quad \mathbf{x} \in \Omega/\Gamma, t \in T, j \in J, \lambda \in \Lambda. \quad (1)$$

By solving, we obtain $\mathbf{u}$, the 3D time-varying gas-phase velocity, pressure, temperature, and species, and $\mathbf{v}$, the Lagrangian particle variables (size, position, velocity, and temperature). Ω , Γ , T , and J represent the microscale domain, boundaries, time span, and the number of Lagrangian particles. In the EL simulations, the governing equations and boundary conditions are

$$\bar{\mathcal{F}}(\bar{\mathbf{u}}(\mathbf{x}, t), \mathbf{v}(j, t), F(j, t), \bar{\lambda}) = 0 \quad \text{and} \quad \bar{\mathcal{B}}(\bar{\mathbf{u}}(\mathbf{x}, F(j, t), \bar{\lambda}) = 0 \quad \text{for} \quad \mathbf{x} \in \bar{\Omega}/\bar{\Gamma}, t \in \bar{T}, j \in \bar{J}, \bar{\lambda} \in \bar{\Lambda}, \quad (2)$$

which are obtained through rigorous filtering (or coarse-graining) of (1) [47]. In the EL approach, the filtered gas-phase fields are $\bar{\mathbf{u}}$, and the particles are unfiltered. $\bar{\Omega}$, $\bar{\Gamma}$, $\bar{T}$, and $\bar{J}$ represent much larger domains and trillions of particles. The filtering operation introduces the closure quantities $F(j, t)$ (force, heat/mass transfer, burn-rate, Reynolds stress, and flux) that must be modeled. In the macroscale EL simulation, each grid cell serves as a *representative volume element* (RVE). The details of group combustion within each RVE remain unresolved and are accounted for through the closure quantities $F(j, t)$. Thus, each RVE is a unique window characterized by the list of parameters λ : local filtered fields and their spatial/temporal gradients (i.e., $\bar{\mathbf{u}}$, $\nabla \bar{\mathbf{u}}$, $\partial \bar{\mathbf{u}}/\partial t$, etc.), and other parameters such as particle and collision properties, represented as $\bar{\lambda}$.

ML-based subgrid closure models are being pursued by a large number of researchers [75–78]. *The proposed approach is unique in several important ways:* (i) we model the subgrid closure terms as a Lagrangian quantity $F(j, t)$ for $j = 1, \dots, J$. We pursue the hypothesis that $F(j, t)$ of the j^{th} particle depends not only on the state of the window (i.e., on λ) but also on the set $I(j, t)$ that includes the state of the N nearest neighbors. (ii) We recognize the presence of unavoidable uncertainties and propose quantifying them by predicting the conditional probability $P(F)$, given I and λ , with the relation $\mathcal{G}(P(F), I, \lambda) = 0$. (iii) The scope of the proposed model far exceeds previous efforts. We intend to learn the conditional probability $P(F)$, for all possible I and $\lambda \in \Lambda$. The best-possible

EL closure can now be mathematically defined as the relation $\mathcal{G}(P(F), I, \lambda) = 0$, which yields the conditional probability $P(F)$ as accurately as in an HF-PR microscale simulation.

(a) HF-PR microscale simulations		(b) PR generative ML surrogate	
$\mathcal{F} = 0, \mathcal{B} = 0$	Eqns. being solved	$\mathcal{F} = 0, \mathcal{B} = 0$	Eqns. being solved
$\{\lambda_n\}_{n=1}^{N_{hf}}$	Input parameters	$\{\mathbf{u}_n, \mathbf{v}_n, \lambda_n\}_{n=1}^{N_{hf}}$	Training data
$\mathcal{D}^{hf} = \{\mathbf{u}_n, \mathbf{v}_n\}_{n=1}^{N_{hf}}$	Primary output	$\mathcal{D}^{lf} = \{\mathbf{u}_n, \mathbf{v}_n, \lambda_n\}_{n=1}^{N_{lf}}$	Primary inference
$\{I_n, P(F_n)\}_{n=1}^{N_{hf}}$	Derived output	$\{I_n, P(F_n)\}_{n=1}^{N_{lf, mic}}$	Derived inference
(c) Operator-learning ML model			
$\mathcal{G}(P(F), I, \lambda) = 0$		Unknown operator being modeled	
$\{I_n, P(F_n), \lambda_n\}_{n=1}^{N_{hf}}, \{I_n, P(F_n), \lambda_n\}_{n=1}^{N_{lf}}$		Multifidelity training data	
$P(F)(j, t)$ given $I(j, t)$ and $\lambda \in \Lambda$		Inference objective	

Table 1: A schematic of the steps required to develop microscale-informed models for EL simulations at the mesoscale via particle-resolved simulations.

Only with many microscale investigations can we properly explore all relevant values of λ to approach the best-possible closure. However, we can afford only a limited number of exascale HF-PR simulations. Thus, the fourth and the most innovative aspect is our two-step approach, where only $N_{hf} \sim O(10^3)$ different values of λ will be considered in the HF-PR simulations. Each HF-PR simulation will yield the computed flow fields as the primary output and I and $P(F)$ as the derived output (see Table 1a). In the first step, we will develop a generative ML model (GMLM) that is trained to solve $\mathcal{F} = 0$ and $\mathcal{B} = 0$ (see §2.5.1) and obtain microscale solutions as in the HF-PR approach. We will use the HF-PR dataset $\mathcal{D}^{hf}$ as the training input. Once trained, owing to its orders of magnitude lower computational cost, the GMLM will be used to sample the parameter space Λ far more extensively ($N_{lf} \gg N_{hf}$) to obtain the surrogate dataset $\mathcal{D}^{lf}$ and the associated derived outputs I and $P(F)$ (blocks 1C and 1D in fig. 6). The surrogate ML predictions will involve additional uncertainties (discussed in §2.3.5). The objective of the second step is to develop operator-learning ML models (OMLM) to learn the operator $\mathcal{G}$ so that best estimates of the closure quantities and their uncertainties can be accurately predicted (block 1E in fig. 6) and used in EL simulations. Table 1 shows that OMLM training uses both the HF-PR simulations and surrogates.

2.3.5 Uncertainty Quantification

In fig. 6, the blue/gray boxes (marked 1B, 1D, 2B, 3A, and 3C) represent the results of micro, meso, and macroscale simulations. They involve uncertainties, which are discussed below:

1. HF microscale results (box 1B) involve (i) epistemic uncertainty arising from reduced-order surface and gas-phase reactions and radiation and (ii) additional uncertainty at higher Reynolds numbers when the particle’s flow and surface characteristics are not fully resolved. Epistemic uncertainty will be quantified by relay validation as outlined above.
2. Microscale surrogate results (box 1D) involve (i) uncertainty in the HF-PR training data propagated through GMLM training, (ii) uncertainty arising from lack of sufficient HF-PR data for proper training, and (iii) epistemic uncertainty arising from deficiencies in GMLM architecture and training protocol. These uncertainties will be quantified as described in §2.7.2.
3. Meso and macroscale EL results (boxes 2B and 3A) involve (i) aleatoric uncertainty arising from lack of precise knowledge of cascading subgrid classical turbulence and neglect of fragmentation/compaction and (ii) propagated uncertainties that arise from three sources of imperfections in the ML-based EL closure models: (iia) uncertainties in the HF and surrogate training data propagated through the OMLM, (iib) uncertainty arising from lack of sufficient training data,

and (iic) epistemic uncertainty arising from deficiencies in OMLM architecture and the training. The quantification of these uncertainties will also be discussed in §2.7.2.

4. Macroscale EE results (box 3c) involve (i) aleatoric uncertainty arising from a lack of knowledge of the particles in the Eulerian representation and (ii) propagated uncertainty from imperfections in the ML-based closure models. These will not be the focus of the proposed work.

2.4 Exascale Plan

The group combustion of explosively dispersing particles is computationally intensive. All aspects of the computation, such as the propagating primary shock, the expanding particles, and the dynamics of interacting flame envelopes, are spatially non-uniform and transitory. The generative and operator-learning ML model training is computationally intensive, with various possible parallelization schemes. The exascale plan aims to develop efficient computational algorithms and software for micro, meso, and macroscale simulations and the training and deployment of ML models. Lightweight operator-learning ML models need to be embedded within the EL simulations as the best-possible closure, and this will require automatic decomposition and load balancing for a mixture of workloads. The proposed work efficiently uses the latest APU “super chip” CPU–GPU single-die architectures, which comprise emerging exascale platforms, including LLNL El Capitan and its CZ sister, Tuolumne.

2.4.1 Demonstrated Exascale Performance on Emerging Architectures

The center is ready to use the exascale compressible multiphase and multiphysics flow code MFC from the outset for all the proposed micro, meso, and macroscale simulations [79]. MFC is an MIT-licensed open-source tool for solving multi-component, compressible multiphase flows with a broad range of modeling capabilities [3, 4, 80, 81]. It can efficiently solve many flows, including droplet atomization, shock–bubble interaction, and bubble dynamics using a 5- or 6-equation thermodynamically consistent diffuse-interface model. The solver employs high-order interface-capturing methods, HLL-type Riemann solvers, and TVD time-integration schemes that can simulate strong shocks. The numerical methods are implemented in a flexible, modular framework amenable to continuous development. The methods employed are validated via comparisons to experimental results for shock–bubble, shock–droplet, and shock–water–cylinder interaction problems and verified to be free of spurious oscillations. The methods are verified to be high-order accurate for smooth solutions. MFC currently computes chemical reactions via coupling to the high-performance library Pyrometheus [82, 83], which is exascale-ready and does on-the-fly code generation.

MFC is currently going through the SPEChpc benchmark onboarding process for its scaling performance and hardware utilization capabilities. MFC offloads all compute kernels to accelerators, including AMD and NVIDIA GPU hardware, via OpenACC compiler directives and achieves better kernel performance than other finite-volume-based solvers. The MFC development team collaborates with the DOE/HPE Cray Center of Excellence on exascale capabilities for optimizing MFC on OLCF Frontier and LLNL’s early access systems Tioga and, soon, Tuolumne. MFC achieves near-optimal performance in part by avoiding many memory-access issues via metaprogramming, enabling otherwise hidden compile-time optimizations. This strategy exposes compile-time optimizations, promotes high memory reuse, and uses remote direct memory access (RDMA) to reduce halo-exchange communication time. Metaprogramming results in an 8-times speedup of the most expensive kernels compared to a statically compiled program, reaching 46% of peak FLOPs on current GPUs and high arithmetic intensity (about 10 FLOPs/byte).

Figure 7 (left) shows the grind time of a representative 3D MFC simulation on various devices and a breakdown of the most expensive compute kernels. All compute kernels are executed by the GPU with no intermediate data exchanges. Routine proportions associated with the different GPU architectures are similar. Other costs include computing source terms, packing halo buffers, and

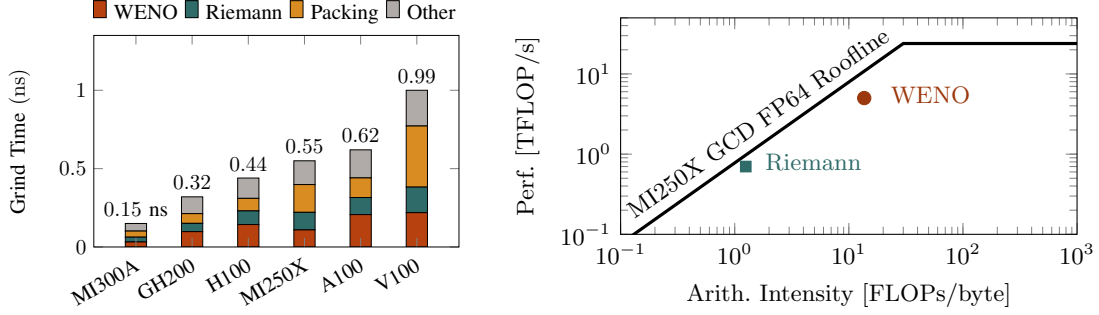


Fig. 7: (Left) Cost breakdown of MFC subroutines on various architectures. All time is associated with GPU compute kernels. The AMD MI300A and MI250X numbers indicate use of the entire device (all XCDs or GCDs). The numbers above the bars indicate the grind time as time per time step, grid point, and right-hand side evaluation. Profiles were collected using `OmniPerf` and `rocProf` (AMD MI300A and MI250X) and `Nsight Systems` and `Nsight Compute` (NVIDIA H/A/V100). (Right) Roofline analysis for the two most expensive compute kernels, the WENO reconstruction and Riemann solve, on OLCF Frontier. All computations are double-precision (FP64) with modified fused multiple-add (MFMA) instructions. MI300A roofline-level profiles are unavailable due to an AMD firmware bug at the time of writing.

other ancillary kernels. For reference, the MI300A grind time is 113 times faster than an Intel Xeon 6226 Cascade Lake CPU (all cores). At the LLNL-spec MI300A node configuration with four such APU units, one expects a 452-times speedup over a single-socket CPU configuration.

Figure 7 (right) shows the performance of the two most expensive kernels in MFC: WENO reconstruction and Riemann solves. We show the roofline assessment of these kernels for the AMD MI250X GPUs. The most expensive kernels are memory bound but still achieve a meaningful fraction of peak FLOPs, especially when compared to three other well-documented compressible CFD solvers [84–86]. For example, the WENO kernel achieves 26% of MI250X peak FP64 MFMA FLOPs. The Riemann kernel is more strongly memory-bound and achieves a smaller fraction of peak FLOPs, but improvements are expected once branching is removed via metaprogramming.

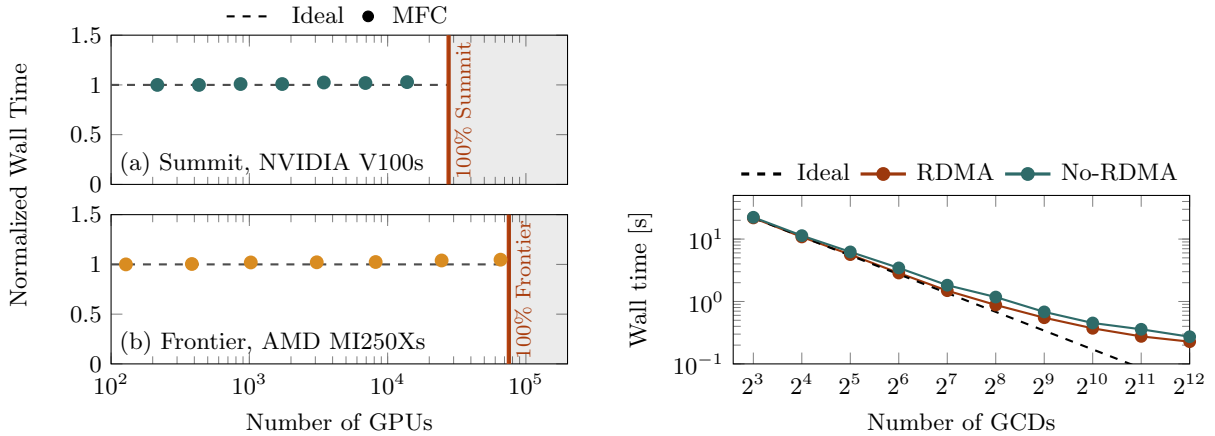


Fig. 8: (Left) Weak scaling of MFC on OLCF Frontier (AMD GPUs) and Summit (NVIDIA GPUs). (Right) MFC strong scaling on OLCF Frontier using standard MPI and GPU-aware MPI via remote-direct memory accesses (RDMA).

Shown in fig. 8 (left), MFC scales to better than 90% of all NVIDIA GPUs on OLCF Summit and 88% of the AMD GPUs (>66K) on OLCF Frontier (the first exascale computer). Each GPU computation is approximately 500 times faster than one on a comparable CPU device. Collective MPI3 I/O helps

ensure the negligible contribution of data transfers ($< 1\%$ of the wall time for a large simulation). **Figure 8** (right) shows strong scaling performance on OLCF Frontier for the same problem as the weak scaling analysis. Full-GPU simulations on the MI250X (32M grid points) show 96% of ideal performance when the GCD count is increased by 16 times. As expected, smaller problems and larger numbers of GPUs lead to worse performance. We use GPU-aware MPI (RDMA) on Frontier via HIP-coupled MPI libraries. This feature further improves our strong scaling performance by nearly 2-times over standard MPI on smaller problems across more GCDs and a larger fraction on intermediate-size simulations (bottom right corners of **fig. 8** (right)).

2.4.2 Proposed Dynamic Load Balancing of Particle-laden Configurations

The optimal execution of the proposed simulations on exascale systems requires close interaction between experts in multiphase flow simulations, numerical algorithms, software development, and high-performance computing. The design of our proposed and implemented algorithms brings together such experts and simultaneously accounts for the diverse features of our flow configurations, focusing on the challenging case of group combustion of explosively dispersing particles.

Exascale architectures are now known, at least in the near term. However, our optimization strategy must remain flexible and account for intra- and inter-node interconnects and network topology variations. The multiphase implementation of the computational framework of MFC has been established. Still, algorithmic variations are available for Lagrangian particle tracking and the coupling between particles (collisions) and with the fluid (interpolation/projection). For example, when conducting EL simulations, one can interpolate Eulerian quantities from the fluid grid to the Lagrangian particles via fast but rather inaccurate linear interpolation or higher-order accurate interpolations consistent with the asymptotic error behavior seen in the gas-phase discretization. Further, we must consider the data structure that holds the Lagrangian particles, which we propose as a dynamically allocatable array at the MPI-rank level. The array can be sorted, traversed quickly, shrunk, or expanded as particles accumulate or dissipate.

The soft-sphere collision algorithm associated with the particles also plays a central role in the computational costs and factors into the computational cost prediction model that balances the workload (discussed further in §2.4.3). We also have many static and dynamic parameters, such as the number of grid points and particles, flux scheme, and chemistry mechanism. Remaining agnostic to this diverse set of features presents a similarly broad range of potentially optimal workloads. Our strategy is to keep a fully dynamic workload manager that pushes memory management to the Zen CPU cores of the MI300A APU devices so that the GPUs are consistently busy. At the same time, the CPU can rearrange memory and balance the workload in a flexible, dynamic manner. This process is CPU-GPU asynchronous since we do not have meaningful all-to-all communication. Overall, the adaptive load-balancing strategy uses the following primary computational and communication factors: the fluid and particle solver costs for each MPI rank, the mapping of grid elements on each MPI rank, and the distribution of the particles among the ranks. The fluid and particle solve costs depend only on their respective local algorithms and the hardware that computes them, though the grid mapping and particle distributions are problem-dependent. The trace-based approach discussed next identifies an appropriate load-balancing strategy for each problem, which can be executed without additional communication overhead.

2.4.3 Trace-based Particle Workload Prediction

Algorithms and parametric details are sufficient to predict the performance of a single-phase flow simulation. So long as the initial condition is known, this diversity of configurations includes exascale systems comprising many MPI ranks and GPU/APU devices. For dispersed multiphase flows, the distribution of Lagrangian particles is typically spatially heterogeneous and time-dependent. When

a time-space discretization is prescribed and mapped on a set of heterogeneous nodes, the particle-phase performance depends on the flow solution. Rather, the simulation grind time of particle phase dynamics depends strongly on the particle distribution, domain decomposition, and local fluid phase. These problems must be considered to achieve a balanced and predictable workload.

We will use a particle-based approach to achieve a balanced workload and enable multiphase reactive simulations that are not meaningfully more expensive than gas-only simulations, even in highly heterogeneous environments across thousands of nodes. The strategy uses the invariance of the particle motion and distribution to numerical and simulation parameters, including the number of grid points, resolution, and XCD saturation. So, a low-fidelity, coarse simulation will be used to generate the *particle trace* with short wall times [87]. Abstracting problem-specific details enables reliable prediction of the computational and communication workload associated with the particles. With these (cheap) predictions, we will use task-based parallelism to schedule the domain decomposition restructuring so that it is asynchronous (again, occurring on the MI300A Zen cores).

2.5 AI/ML Plan

The success of the ambitious multiscale framework presented in [fig. 6](#) relies on our ability to develop the best-possible closure models, which, as defined in [§2.3.4](#), must predict the subgrid quantities and their uncertainties under all possible microscale configurations that are likely to occur in the demonstration problem. Until recently, this goal has remained unreachable since it requires that the model learn to capture the complete conditional statistics of the closure relation $\mathcal{G}(P(F), I, \lambda)$ in its entirety. Recent ML models have demonstrated an uncanny ability to capture complex multi-dimensional statistical distributions of vast parametric dependence. We plan to exploit these developments with a novel approach to bridge the scales with the best-possible closure models.

As discussed earlier, our two-step approach is the key aspect of the proposed melding of SC and ML. As represented by blocks $1B \rightarrow 1C \rightarrow 1D$ in [fig. 6](#), the first step is to develop a GMLM that is trained to solve the governing equations (see [table 1](#)). The trained model will then obtain orders of magnitude more surrogate solutions, spanning the probabilistic space more efficiently than HF-PR simulations. In the second step, represented by blocks $1B \rightarrow 1E \leftarrow 1D$, we develop the OMLM that learns the relation $\mathcal{G}(P(F), I, \lambda) = 0$ from the HF-PR and surrogate data. It is natural to ask: *Can the best-possible OMLM be derived from the HF-PR data directly, in one step?* If GMLM can generate accurate surrogate data, couldn't an OMLM be trained using the same data directly?

The answer is “No.” The fundamental difference is in data availability. Each of the $N_{hf} \sim O(10^2)$ exascale HF-PR simulations will yield highly-resolved results of group combustion around $N_p \sim O(10^3)$ particles. These simulations yield a derived dataset of size $N_{hf}N_p \sim O(10^5)$ for the closure term F . Since the neighborhood of each particle is resolved with a fine grid of size $N_g \sim O(100)$, the HF-PR simulations yield a much richer database of reactive flow information at $N_{hf}N_pN_g^3 \sim O(10^{11})$ grid points. Thus, with the limited HF-PR simulations, we have a substantial amount of reactive flow information to train the GMLM but less closure information to train the OMLMs. Another important advantage favors GMLM. In the training of GMLM, we pursue a hybrid approach where the governing equations are imposed as loss functions in addition to supervised training loss [88–90]. With a physics-informed neural network, an accurate GMLM can be constructed even without *any* training data [91–94] but at significantly added training cost, which we attempt to avoid with the hybrid approach. This strategy is impossible with the OMLMs since we are attempting to learn unknown closure relations. The advantage of first training a GMLM over directly training subgrid closure models can be inferred in recent multiphase studies [95–98].

Another important innovation is how we plan to accurately capture the deterministic and stochastic effects of the unresolved scales $\mathbf{u}' = \mathbf{u} - \bar{\mathbf{u}}$ on the resolved scale dynamics. To facilitate modeling,

we conceptually separate the unresolved scales into three contributions $\mathbf{u}' = \mathbf{u}'_{dt,ps} + \mathbf{u}'_{st,ps} + \mathbf{u}'_{ct}$, where $\mathbf{u}'_{dt,ps} + \mathbf{u}'_{st,ps}$ accounts for pseudo turbulence. The (dt) term accounts for pseudo turbulence that is deterministically dependent on the particles, while the (st) term accounts for the stochastic component expected at higher Re . The unresolved flow due to classical cascading turbulence is $\mathbf{u}'_{ct}$. The strategy for modeling the deterministic component $\mathbf{u}'_{dt,ps}$ and its effect on closure terms F_{dt} will be discussed in the first two subsections. The strategy for modeling the stochastic component $\mathbf{u}'_{st} = \mathbf{u}'_{st,ps} + \mathbf{u}'_{ct}$ and its effect of closure terms F_{st} will be discussed in §2.5.3.

2.5.1 Generative ML Models

We define generative ML models as neural networks (NN) that are capable of solving systems of PDEs. Their attraction is their ability to obtain accurate solutions at orders of magnitude lower computational cost [75, 76, 99–101]. Of particular relevance are the generative adversarial network (GAN) [102, 103], denoising diffusion probabilistic models (DDPM) [104, 105], and deep nonlinear system identification [106–108] that can be trained to predict the microscale state of a random distribution of particles [95].

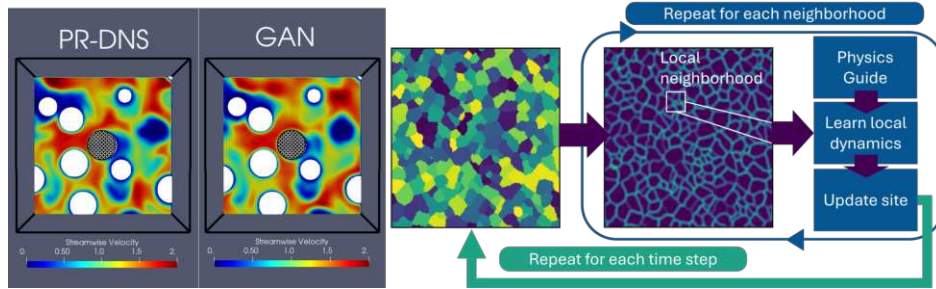


Fig. 9: Generative ML. Left: (PR-DNS) A small vertical section of HF-PR velocity around a reference particle (hatched) surrounded by other particles within an LVE. Each HF-PR simulation involving $O(1000)$ particles takes 4000 CPU hours. (GAN) Surrogate generated by a trained GAN. GAN takes only 0.39s per particle on one GPU. Right: PRIMME architecture for learning micro-structure of grain growth dynamics, utilizing a nonlinear system identification strategy. After the neural network learn the “local dynamics”, the learnt behavior of the grains can be applied to any arbitrarily sized initial domain to generate grain growth.

GAN has been successfully used to recast the flow prediction problem from the perspective of an individual particle [95, 109] (fig. 9 (left)). The GAN was trained to solve the governing equations within a small local volume element (LVE) around a reference particle (hatched) with the knowledge of the immediate neighbors (white). This has the advantages of (i) increasing the training data to $N_{hf}N_p$ particles and (ii) reducing the ML prediction to be only within the LVE ($\ll$ RVE).

Here, we propose using deep nonlinear system identification to serve as a hybrid approach that combines the advantages of supervised and unsupervised approaches [106–108]. The network will be trained to solve the governing equations at a single grid point with the knowledge of its location with respect to nearby particles. The advantages are far greater: (i) the training data explodes to $N_{hf}N_pN_g^3$, and (ii) the scope of prediction shrinks to a single grid point. Furthermore, time evolution and physics regularization (e.g., governing equations) can be imposed before training, and help guide solutions during and after training. Such an approach has been applied to polycrystal grain boundary evolution [106, 107] (see fig. 9 (right)).

2.5.2 Graph Convolution Networks for N-body Operator Learning

We now turn to OMLM, which attempts to model the relation $\mathcal{G}$ by predicting the conditional distribution $P(F)$ of the closure quantities of a particle given the microscale details of the particle’s immediate neighbors (i.e., I) and the local macroscale state (i.e. λ). By posing the closure problem in this Lagrangian manner, we reduce it to the classic N -body problem encountered in many branches

of physics, such as molecular dynamics. However, in the present problem of group combustion, we are faced with modeling the interaction of flame envelopes and associated pseudo turbulence. The challenge to developing such an N -body group combustion model is the curse of dimensionality that arises from the huge parametric dependence. For example, the burn rate of a particle modeled as a function of its N nearest neighbors must depend on $O(N)$ vectors and scalars that represent the neighbor's relative position, velocity, temperature, burn-rate, and more.

We propose to use graph convolutional networks (GCN) to model this closure of N -body group combustion. GCNs are well-suited for modeling N -body interactions as sums of independent pairwise influences to maximize the likelihood of accurate representation among all possible multivariate distributions [110, 111]. In group combustion, a few immediate neighbors most strongly influence the burn rate of a particle, and the neighbors, in turn, are influenced by their neighbors, and so on [112, 113]. Correspondingly, in a GCN, the edges to a node account for the node's *local* interactions with its immediate neighbors, while the *global* influences are captured by the graph as a whole. In calculating the influence of neighbors, it should not matter how the neighbors are ordered and how their influences are aggregated. In other words, the model must be permutation invariant. Furthermore, from physical principles, it can be argued that the model must universally apply to all the particles. The convolutional nature of GCN satisfies these discrete invariance principles. Because of these advantages, GCN is ideal to model N -body problems [83, 114–116].

Our architectural design will benefit from three other theoretical considerations: (i) *Hierarchical expansion*: We will exploit a hierarchical expansion of the N -body problem as a superposition of unary, binary, and trinary interactions with loop, edge, and face connections [83, 97, 98, 115, 116]. (ii) *Method of reflection*: is an iterative approach that allows a fully coupled solution of an N -body problem through a sequence of binary coupled solutions. In a GCN, its layers implement the method of reflection. *Modal expansion*: We separate the N -body prediction into a sequence of interpretable eigenmodes that depend on the vector parameters and learnable modal amplitudes that depend on the scalar parameters, which can be implemented as two sub-networks. With these important considerations, the model form of a closure quantity F of the α^{th} particle can be expressed as

$$F^\alpha(I, \lambda) = F^1(\lambda) + \sum_{\beta=1}^N ' \left\{ \underbrace{\sum_{l=1}^{\infty} \psi^{2,l}(\mathbf{e}^\alpha, \mathbf{e}^\beta, \mathbf{x}^{\alpha,\beta})}_{\text{binary modes}} \underbrace{A^{2,l}(|\mathbf{V}^\alpha|, |\mathbf{V}^\beta|, \lambda)}_{\text{binary amplitudes}} \right\} + \text{trinary term} + \dots \quad (3)$$

In this representation, the binary contribution from the β^{th} neighbor is expressed as a weighted add-mixture of different spatial modes. The scalar and vector dependencies have been separated. The binary modes depend on the separation vector $\mathbf{x}^{\alpha,\beta}$. The symmetry of this dependence is dictated by the direction of their relative velocity with respect to the ambient flow (i.e., $\mathbf{e}^\alpha$ and $\mathbf{e}^\beta$). The modal amplitudes depend on the relative velocity magnitudes and the set λ .

A sample three-layer GCN of the binary term is shown in fig. 10. We assume that each particle is influenced by its M_b closest neighbors (i.e., each node is connected to M_b other nodes). The

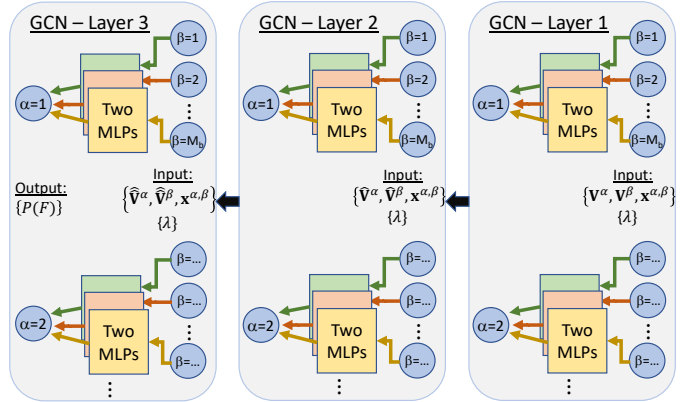


Fig. 10: Three-layer binary GCN.

graph’s edges are directed, and the adjacency matrix is not symmetric. The inter-particle separation vector characterizes each directed edge $\{\mathbf{x}^{\alpha,\beta}\}$ as the feature. For each edge-connected neighbor of a particle, its nodal, edge, and global features are combined to form the input. The output of all the M_b neighbors is then pooled together through an aggregation process to form the final first-layer output for each particle in the system. The key interpretation of the latent state of the first-layer is that its output is the first reflection at the α^{th} particle.

2.5.3 Mori–Zwanzig Formalism for reduced-order Models

The past two subsections provide a path to modeling the deterministic effect of particle-generated group combustion from the precise knowledge of the particles within the RVE. We adopt Mori and Zwanzig’s (MZ) functional projection approach for the stochastic contribution. A pre-specified projection operator is used to map the stochastic portion of closure quantities $P(F_{st})$ that depends on $\mathbf{u}'_{st}$. We do not need the precise knowledge of the Lagrangian particles for this closure. Only averaged information, such as local volume fraction and heat release, is sufficient. The central result of the MZ formalism is the Generalized Langevin Equation (GLE), which stipulates that the evolutionary equations of the resolved variables depend not only on their instantaneous values but also their history. The MZ formalism further quantifies the uncertainty and derives how the uncertainty would propagate into the future.

There are many choices of the projection operators: (i) Mori’s linear projection [117], (ii) finite-rank projection [118], (iii) projection by assigning unresolved variables to zero [119–122], and (iv) Zwanzig’s fully nonlinear projection [118, 123]. Recent studies [124–130] have established that with Mori’s linear projection it is possible to learn the MZ operators using the time series of the resolved dynamics. These methods provide higher-order and memory-dependent corrections to existing data-driven learning of the approximate Koopman operators [129, 131]. For a discretized system given by $\mathbf{u}(t+1) = \Phi(\mathbf{u}(t))$, where $\Phi: \mathbb{R}^D \rightarrow \mathbb{R}^D$ is the flow map, an exact evolutionary equation can be derived for the closure quantities as [129, 132]:

$$F_{st}^{(n+1)} = \sum_{\ell=0}^n \Omega^{(\ell)} F_{st}^{(n-\ell)} + \mathbf{W}^{(n)}. \quad (4)$$

The operators $\Omega^{(\ell)}$ contain the Markov term ($\ell = 0$) and additional memory effects ($\ell > 1$), while $\mathbf{W}^{(n)}$ represents the orthogonal dynamics (i.e., representing error or noise) and contains all information not captured. The Generalized Fluctuation-Dissipation (GFD) theorem relates the memory kernel and the error and can be stated for the discrete representation as $\Omega^{(n)} F = \mathcal{P} \mathcal{K} \mathbf{W}^{(n-1)}$, $\forall n \in \mathbb{N}$, where $\mathcal{P}$ and $\mathcal{K}$ are the projection and Koopman operators, respectively. The memory kernels of MZ formalism are special because they satisfy the GFD. Other ML models, such as recurrent NN with long short-term memory (RNN-LSTM), do not enforce GFD [129]. For large systems such as the proposed problem, GFD is needed to establish a recursive procedure for learning the memory kernel and extracting it computationally efficiently. The MZ formalism is needed only to predict the stochastic component of the closure terms. Thus, it will be used in conjunction with the GCN, as there is no need for a generative model to extract the stochastic component $\mathbf{u}'_{st}$.

2.6 Software Plan

2.6.1 Simulation & Modeling Roadmap

Bryngelson will be the center’s Chief Software Architect (CSA) and maintainer. In addition to simulating the demonstration problem, we will perform increasingly complex simulations at the micro and mesoscales. The ML models developed from these micro/mesoscale simulations will be used in Y3–Y5 macroscale simulations. To take advantage of validation experiments, many low-fidelity Euler-Euler simulations will be used for calibration.

We will start with GPU computers with the AMD MI250X and NVIDIA GPUs at DOE and other facilities. APU indicates the AMD MI300A CPU–GPU units on LLNL El Capitan and NVIDIA GraceHopper-like superchips as they become available. MPI ranks are associated with the total GCDs and XCDs on Lassen and El Capitan/Tuolumne. The grid and the number of variables per cell increase in size from Y1–Y5 via mixed-precision schemes and on-the-fly compression. The number of particles is typically on the order of the mesh size. We will conduct annual hero simulations at very high resolution on the latest computer made available to us. MFC already scales ideally ~100% of OLCF Frontier (AMD MI250X) and Summit (NV V100) and is performant on single node instances of LLNL LC Tioga (AMD MI300A, El Capitan early access); see §2.4.1 for further details on scaling and performance.

	Y1	Y2	Y3	Y4	Y5
Gas-phase physics and MFC development					
Gas phase	LES	LES	LES	LES	LES
EOS	Ideal	JWL	BKW	BKW	BKW
Closure model	Standard Smag.	Dynamic Smag.	Dynamic Smag.	ML-based	ML-based
Chem. reactions	$\mathcal{O}(1)$	$\mathcal{O}(10)$	$\mathcal{O}(10)$	$\mathcal{O}(100)$	$\mathcal{O}(100)$
Wall effects	Inviscid	Inviscid	Inviscid	Inviscid	Viscous
Grid size	10^{12}	10^{12}	10^{12}	10^{12}	10^{13}
Euler–Lagrange MFC developments and details					
Dispersion (EL)	Particles: C	Particles: C	Particles: C	Particles: C	Particles: C
	Droplets: Mg	Droplets: Mg	Droplets: Mg	Droplets: Mg	Droplets: Mg
Concentration	Dilute	Dilute	Dense	Dense	Dense
Contact model	None	Soft sphere	Frictional	Frictional	Frictional
			soft sphere	soft sphere	soft sphere
Force model	Standard	Standard	ML-based	ML-based	ML-based
Particles	10^{10}	10^{11}	10^{12}	10^{13}	10^{13}
Load balancing	Simple	Bin-based	Bin-based	Trace-based	Trace-based
Hero Sims (EL)	2	2	4	4	4
Euler–Euler MFC developments and details					
Dispersion (EE)	Particles: C	Particles: C	Particles: C	Particles: C	Particles: C
	Droplets: Mg	Droplets: Mg	Droplets: Mg	Droplets: Mg	Droplets: Mg
Distribution	Monodisperse	Polydisperse	Polydisperse	Polydisperse	Polydisperse
Grid size	10^{10}	10^{10}	10^{10}	10^{10}	10^{10}
Grid refinement	Static	Static	AMR	AMR	AMR
Concentration	Dilute–dense	Dilute–dense	Dilute–dense	Dilute–dense	Dilute–dense
Contact model	Granular	Granular	Granular	ML-based	ML-based
Force model	Standard	Standard	ML-based	ML-based	ML-based
EE Sims.	50	100	300	600	1000
Exascale plan					
Architecture	GPU (Lassen)	APU	APU	APU	APU
	APU (El Capitan)	(El Capitan)	(El Capitan)	(El Capitan)	(El Capitan)
Machine utilization	~100%	~100%	~100%	~100%	~100%

Table 2: The modeling and simulation roadmap for the demonstration problem. The roadmap shows progressively more complex simulations for the five years. ML: machine learning, EOS: equation of state, LES: large eddy simulation, JWL: Jones–Wilkins–Lee, BKW: Becker–Kistiakowsky–Wilson, Smag.: Smagorinsky.

The existing MFC solver is documented in the first open-source release [3], its companion papers and website [4, 80, 81, 133]. Documentation is maintained of new features and their performance and correctness. Previous section §2.4.1 describes existing features of MFC. They include multi-component models via a diffuse interface capturing approach with support for full-resolution multiphase mixtures, phase change, surface tension, shock and interface capturing, subgrid physics

for polydisperse two-way coupled particles/droplets/bubbles with EE and EL descriptions, simple thermochemistry, and more. Algorithmic improvements to the existing solvers will accomplish the macroscale simulation roadmap and companion micro/mesoscale simulations. They include efficient pre-conditioned implementations of detailed stiff chemistry, soft sphere discrete-element collisions, and Euler-Lagrange load balancing.

2.6.2 Lagrangian Particle Tracking, Collision, and their Balancing

Key to the proposed explosive dispersal simulations is four-way coupled Lagrangian particle tracking. Such tracking enables the accurate accounting of collisions. Particles may not be uniformly distributed over the flow volume at any given time, and localized particle clusters can enhance the collision algorithm’s cost. These challenges are configuration-dependent, and a universal strategy that performs well in as many such configurations as possible is desired. Here, we extend MFC capabilities to include a bin-based ghost particle algorithm, which enables an exascale implementation of back-coupling from the particles to the Eulerian flow and inter-particle interactions. A bin-based mapping strategy has been shown to preserve the fluid solver’s ideal weak scaling [134–137].

We will pursue two strategies for tackling the particle balancing problem. The first is fluid-based, wherein we ensure that the fluid and particles of the same element are within the same local memory, so no memory transfers are required. Dynamic load balancing is achieved by remapping an uneven number of grid points to the different MPI ranks so that the flow-particle workload is balanced among them. This approach will not be optimal for cases with extremely heterogeneous particle distributions. To address this limitation, we will pursue a bin-based methodology. Here, the particles are uniformly distributed among the MPI ranks and independent of the Eulerian grid distribution. Thus, both the fluid and particle are load-balanced but at the cost of communication during interpolation and projection. We will still dynamically distribute the particles as a compromise against inter-node communication. This communication penalty will be enforced dynamically as inter-node topology, the balance between network latency and bandwidth, and the relative cost imbalance cannot be fully known *a priori*. The dynamic rebalancing work will be done by the Zen CPU cores of the MI300A device, which are more appropriate for handling memory communication problems. The GPU will be free to compute the more compute-intensive fluid and particle kernels.

2.6.3 Other Software Advancements

Data structures: Profiles show that a significant portion of each time step is spent converting struct-of-array data to simple 4D arrays and permuting array indices to facilitate access at different stages of each time step. These unpacks and permutations are currently implemented with fully collapsed three or four-loop OpenACC kernels, organized for coalesced access to the lowest rank of the destination array. More efficient transpose algorithms exist for GPUs, but the current best practice is to use the BLAS extension **GEAM**. NVIDIA originated this extension in **cuBLAS** but only implements a **cublas<T>geam** that can perform a single 2D transpose. AMD’s **hipBLAS** library, part of the ROCm software stack, provides a strided, batch double-precision **GEAM** operation that can efficiently transpose the first two dimensions of a 3D array. These library routines implement optimal algorithms for the AMD hardware and remove the need to optimize for new GPU generations. The conversion to library-based data reshaping reduced computational time in these kernels by 7 times, and they now take a negligible amount of wall time. Direct memory access: We use the remote direct memory access (RDMA, GPU-aware MPI) capabilities of LLNL Lassen and Tuolumne hardware. RDMA via HIP-based ROCm libraries leads to at least twice as fast transfers of halo regions. While the MPI proportion of our production simulation times is not especially large, this feature still benefits our strong scaling performance. On- and near-node NVMe storage: We use file-per-process I/O dumps combined with intermediate data storage on the El Capitan-style node NVMe drives. This strategy is at least 10-times faster than file-per-timestep I/O (without NVMe) when testing

on Frontier. For example, for simulations on $> 80\%$ of Frontier, we use NVMe storage to perform simulations without causing system-wide file system issues.

2.6.4 Code Verification

MFC has already been verified on a broad set of multiphase problems involving shockwaves and high-density ratios. Still, new physics will be introduced during the performance period, requiring careful verification. We use the standard code verification process that involves analytical solution tests, code-code comparisons (against HyBurn [8, 41–43, 138–140] calculations such as ones shown in figs. 3 and 4.), discretization error quantification, and convergence analysis. The method of manufactured solutions is fundamental, ensuring the stability properties of the gas and particle solvers. Verification tests are part of the MFC test suite run via continuous integration, and each new feature must be bolstered by its own set of supporting code verification tests. Ensuring that center members have tested their code is readily discovered via CodeCov, a tool integrated into MFC’s CI that automatically shows what lines of code are changed and which lines are tested. One challenge in verifying multiphase simulation is that the standard EL approach does not show grid convergence, as the grid size approaches the particle size due to self-induced perturbation. Novel correction techniques have recently emerged [141–149]. We will extend these techniques to reactive particles, thereby developing a rigorous grid convergence approach.

2.6.5 Software Design

MFC is an open-source Fortran codebase with a Python frontend. The MFC algorithm stores and uses coalesced memory arrays for stencil reconstructions. This strategy leads to the relatively high arithmetic intensity and performant code on GPU accelerators. We employ a directional splitting approach that evaluates these quantities once per time step, then reuses them, a state-of-the-art strategy for high-order accurate finite volume reconstruction [3, 150, 151]. MFC uses the well-established HDF5, Silo, and ZFP libraries [152–154] for I/O, data storage, and compression, and a metaprogramming strategy via Fypp [155] for case-specific debranching of the MFC runtime and more performant code. Otherwise, MFC employs established methods, including high-order accurate interface reconstruction, approximate Riemann solvers, and TVD Runge-Kutta time stepping.

Parallel processing is handled via MPI3 and domain decomposition. GPU offloading is accomplished via OpenACC. However, we *do not* rely on advanced, new, or NVIDIA-compiler-specific OpenACC features like deep-copies, so we only require OpenACC 2.0 specification and select parts of 2.6. Thus, MFC is performant on AMD GPUs using *current* HPE CCE compilers. Our close collaborations with OLCF researchers and CCE staff engineers enabled this performance portability. We worked with Spock and Crusher before Frontier was deployed, and now Tioga before the deployment of El Capitan. We also participated in two Frontier Hackathons to optimize performance holistically.

The MFC sub-codes were designed to separate the role of analysis. We can restart simulations on-demand via checkpointing. Execution of these sub-codes is automated via Python scripts. These tools operate efficiently at scale, reading output files in parallel on demand. Their cost is low, so we do not expect bottlenecks in checkpointing regimes. In the proposed problem, only infrequent saves are required; therefore, I/O overhead is insignificant. We output Silo and ZFP compressed HDF5 files for parallel visualization using VisIt and Paraview. An analysis is readily available via the MFC post-processing code. Level-3 MPI parallel I/O handles the data flow efficiently and at scale.

2.6.6 Testing and New Developments

MFC has been developed according to strict design rules and with a consistent philosophy and style. Any new routines and modules will follow the same philosophy and style. Bryngelson will oversee MFC’s overall development. The software team will meet biweekly to provide status updates, discuss problems, and seek solutions. MFC also commits to the below standards.

1. **Version control and defect tracking.** Students and postdocs commit changes to their personal forks and issue pull requests (PRs) to the central repository. Once these changes have passed all ~250 regression tests, PRs are performance benchmarked for regressions using systems with the same architectures at Livermore Computing systems. MFC is tested and benchmarked on *every* GitHub push via CI and the MFC test suite, which serves runners for NVIDIA and AMD GPUs, Intel and AMD CPUs, and Cray, Intel, GNU, and NVIDIA compilers. Errors, bugs, and other defects are flagged via GitHub issues at the community and center level.
2. **Addition of new code features and enhancements.** We will continue using GitHub PRs to bring new features into the code. Each PR requires the approval of at least two central code maintainers (co-PIs Bryngelson and Hoium). Monthly releases are scheduled to provide touchpoints for the center broadly.
3. **Documentation & Tutorials.** Each component of MFC has user, developer, and verification and validation guides. These are deployed via GitHub pages using standard Markdown and can be viewed on the MFC documentation webpage or deployed as PDFs. API documentation is automatically generated for all sub-codes via Doxygen. All center employees will document and be responsible for the feature or tutorial they add to MFC. Documentation is automatically deployed via CD (continuous deployment) on GitHub from the same MFC repository and the main codebase. The user guide will also contain tutorials.
4. **Reproducibility & Regression Tests.** The results produced from the simulations will be many petabytes. Instead, when appropriate, input files and post-processing scripts for production runs will be placed in an archival directory for future reference. This enables simulation cases to be used throughout the project as non-trivial tutorials. Other researchers will be able to, trivially, reproduce our results. MFC’s test suite will be extended as new features are added. The test suite will consist of a hierarchy of test cases organized into levels ranging from small to larger 3D tests. A test is judged to have failed if test-specific metrics do not binary-match those of a reference run stored in a repository. No features will be merged until they pass all tests.

2.7 Integration Plan

Tight integration between the physics, VVUQ, exascale, AI/ML, and software plans is essential. We propose to implement the following integration plans: (i) Hierarchical uncertainty propagation that integrates experiments and simulations across the different campaigns; (ii) Uncertainty reduction through active learning (iii) Integration of exascale simulations and ML models; (iv) Software-level integration. (iv) Data integration and workflow for exascale performance.

2.7.1 Integration of UQ and UP Across Scales

Figure 11 shows the hierarchical integration of experiments, HF-PR simulations, EL simulations, and ML models. The relay validation from carbon rods to C/Mg particles, UP across the different scales, and feedback to lower scales for UR are depicted. At the bottom of the uncertainty pyramid, we start with the microscale simulations and experiments of random arrangements of burning carbon rods subjected to high-speed cross-flow. The experiments include (i) sampling uncertainty, (ii) measurement errors, and (iii) measurement processing uncertainties. The UQ in the companion microscale simulations was presented in §2.3.5. For UR, repeated experiments, high-fidelity measurements, increasing resolution in the simulations, and more detailed chemistry will be used. The key outcome is UQ of reduced-order chemistry of carbon group combustion, which will be propagated to the microscale simulations of carbon particles.

HF-PR simulations of carbon particles will be used to train the GMLM as outlined in §2.5.1. The uncertainties associated with the HF-PR simulations will have already been assessed with the rod configuration. By comparing the *test* HF-PR simulations (not used in the GMLM training) against the corresponding GMLM surrogates we will establish the additional uncertainties incurred in

GMLM, which were listed in §2.3.5. The HF-PR simulation and surrogate results will train the OMLM. As pointed out in §2.3.5, there are additional uncertainties associated with the architecture and training of OMLM. We will compute the EL closure quantities from *test* HF-PR simulations (not used in the OMLM training) to serve as benchmark, which will be compared with the corresponding OMLM predictions to establish the uncertainties associated with the closure models.

The OMLM will serve as the best-possible closure in the EL simulations of the two mesoscale problems and the demonstration problem. As pointed out in §2.3.5, additional uncertainties are incurred in the EL simulations. The mesoscale and demonstration experiments will help validate the best-possible EL simulation framework. Again, it is essential to contain the uncertainties in the first- and second-order prediction metrics obtained from the experiments and simulations. We will be engaged in vigorous uncertainty reduction (see §2.7.2). As the final step, the OMLM EL closures will be averaged over all possible particle distributions to obtain the EE closure models. This coarse-graining from EL to EE involves additional uncertainties, which will remain future work to be pursued. By limiting the microscale experiments to only carbon rods, the above UQ workflow applies only for the C particles. This understanding must be leveraged to the UQ of Mg particles.

2.7.2 Uncertainty Reduction - Symmetries & Active Learning

With many contributing uncertainties, there is a great danger that prediction may become meaningless. Therefore, we propose reducing uncertainties using the uncertainty budget (UB) [156–158]. The inherent variability (aleatoric uncertainty) is irreducible, which we will quantify with characterization and calibration of the input. The epistemic uncertainties will be reduced with increased HF data and more accurate generative and operator-learning models. Epistemic uncertainties will be quantified and reduced through Bayesian techniques. This involves defining model likelihood conditioned on inputs and trainable neural network (NN) parameters. Then inference requires an expensive integration over probability distributions [159]. Since this does not scale well, we will sample each parameter of λ independently from a Gaussian distribution or using Monte Carlo dropout during training and inference. One approach to quantifying the ML model’s contribution to uncertainty is to train an ensemble of NNs [160, 161].

Uncertainty in OMLM will also be reduced by (i) enforcing all the physical constraints, symmetries, and temporal correlations in the network architecture and (ii) enhancing the availability and accuracy of training data. The mapping between the input and output features to be captured by the ML models must obey underlying symmetries, such as translational/Galilean invariances and rotational/reflectional equivariances. Novel ML networks have recently been developed in which the symmetries have been baked-in [113, 162–166]. Generative and operator-learning ML networks with built-in rotational and reflectional symmetries [97, 109, 167–169] and with hard imposition of kinematic conditions [170, 171] have been shown to increase prediction accuracy under data-limited conditions. Proper choice of parametric combinations $\{\lambda_n\}_{n=1}^{N_{hf}}$ (table 1 (a))

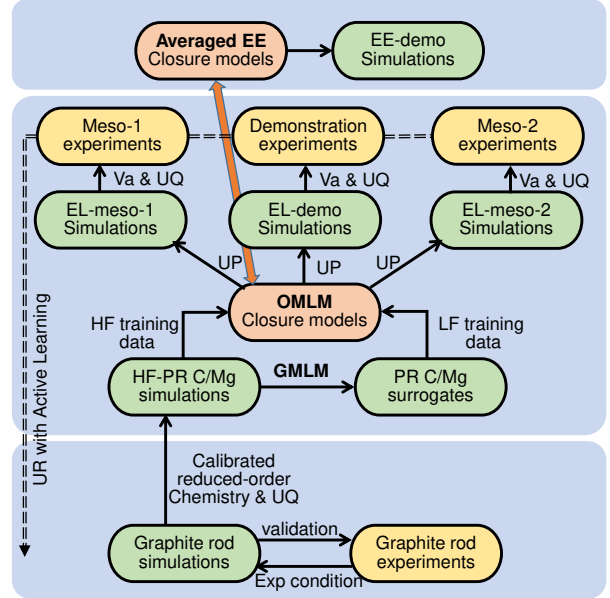


Fig. 11: Hierarchical validation (Va), uncertainty quantification (UQ), uncertainty propagation (UP), and uncertainty reduction (UR) plan.

and $\{\lambda_n\}_{n=1}^{N_{lf}}$ (table 1 (b)) to be used in training the OMLM is important. Their initial choice will employ the design of experiments [49, 172, 173]. Since these will be performed over a period of time, uncertainty reduction will dictate the parametric combinations to be considered in each successive set. Active learner strategy will be used to identify the parametric candidates for the next set of HF-PR simulations to minimize uncertainty [174–177]. The MZ formulation will help as the convergence of operators indicates sufficiency of data coverage [130].

Bayesian inference will calibrate the scale bridging of the generative and operator-learning models. If the level of uncertainty is not acceptable, then UB, as presented in fig. 12, is used to determine the most effective way of reducing uncertainty. Global sensitivity analysis will be utilized to rank the importance of different uncertainty sources contributing to the QoIs. Considering the contribution of individual uncertainties and the difficulty of reducing them, UR should be performed iteratively until the accumulated uncertainty is small enough to reveal the model error. If the model error is large, then model improvement will be performed to improve the prediction capability. This process must be repeated until the model error is reduced to an acceptable level.

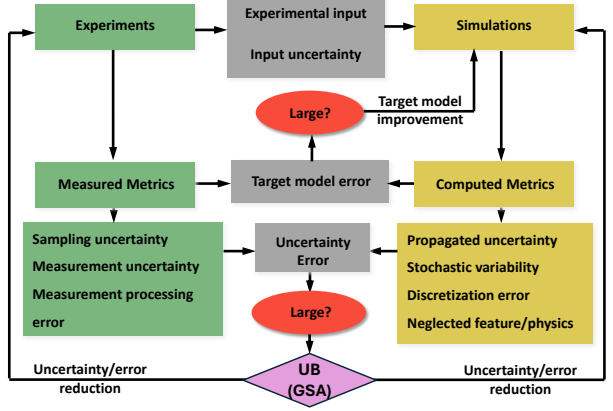


Fig. 12: UB workflow to be pursued at all scales.

2.7.3 Integration of Exascale Simulations and ML models

We plan to achieve unprecedented super-resolution by embedding the best-possible closure models within rigorously derived EL equations through the following six steps (see fig. 13). (1) Exascale HF-PR simulations: The first step is to perform exascale particle-resolved microscale simulations of group combustion under conditions of relevance to the demonstration problem. We envision each of these exascale DNS to contain about 10^3 fully-resolved particles at a total resolution of $O(10^{11})$ grid points. (2) Curation of training data: The DNS data will be processed to obtain local time-resolved flow around each of the particles in a frame moving with that particle, along with the dynamics of its neighbors. The number of particles in each HF-PR DNS determines the number of extracted training data. Each 3D training data set is expected to be 400^3 , and the number of time instances will be $O(10^4)$. (3) GMLM training: The purpose of the next two steps is to generate at least three orders of magnitude more surrogate data that accurately mimic the HF-PR results of Step 1 but at a fraction of the computational cost. Important physical requirements will constrain the training process. (4) Deployment of GMLM: The trained model will generate surrogate 3D time-resolved flow fields around a random distribution of burning particles for a wide range of input parameters that span Λ . It is important to note that each generated surrogate flow will be at a grid resolution of 400^3 , and the number of time instances will be $O(10^4)$. (5) OMLM training: The results of steps 2 and 4 will form the training data, and some of the curated HF-PR results will be reserved for optimizing hyper-parameters as well as testing and uncertainty quantification. This step will train the learnable weights of the multilayer GCN described in fig. 10 to generate lightweight closure models. (6) EL simulations of demonstration problem: The trained lightweight OMLM models will be used as the best-possible closures in the EL simulations of the demonstration problem. The results will then be validated against the companion experiments.

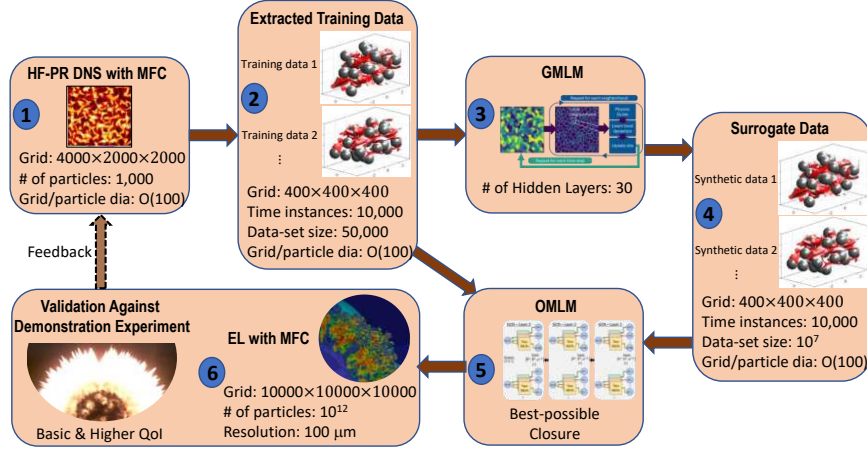


Fig. 13: Integration of simulations and ML models.

2.7.4 Software-level integration

The rise of neural network-based artificial intelligence ushered in high-level libraries, including TensorFlow and PyTorch, to support their functionality. We have benefited from this trend and produced powerful neural networks that promise shorter simulation times. For example, multilayer perceptrons (MLPs) and Long Short Term Memory (LSTM) graph-based architectures can represent subgrid physical effects. Implementing neural networks in CFD solvers is challenging because the programming languages used for artificial intelligence and CFD are mostly non-overlapping. We will use the RoseNNA library, developed by Bryngelson’s group [178], to bridge the gap between neural network inference and CFD.

RoseNNA is a noninvasive, lightweight, and performant tool for ML inference. It focuses on the smaller networks used to augment CFD solvers written in C/C++ or Fortran. RoseNNA accomplishes this by automatically converting trained models from typical neural network training packages into a high-performance Fortran library with C and Fortran APIs. This reduces the effort needed to access trained neural networks and maintains the performance of the CFD solvers we build and rely upon. RoseNNA reliably outperforms PyTorch (Python) and libtorch (C++) on MLPs with less than 100 hidden layers and 100 neurons per layer, even after removing the overhead cost of API calls. Speedups here are about a factor of 10 faster for the RoseNNA library.

2.7.5 Workflow and Data Integration

Our experience with center-level research has shown that the greatest challenge to success is the presence of bottlenecks in the flow of information and data between the different steps of an integration cycle. Without a well-thought-out workflow and data integration plan, even small miscommunications result in weeks of wasted time in debugging and disagreement. To address this problem, we focus on streamlining the workflow and data exchange between the various participating teams in executing the above integration plans. The workflow/data management will be enforced by a designated workflow manager (WM) with expertise [179–183].

PSAAP centers face a unique challenge: the UQ effort will be carried out in a dynamic environment where the simulation software, numerical algorithms, and physical models are iteratively being upgraded. In such an environment, it is common to see different versions of codes and results under different models and input conditions. This can be most detrimental to UQ and validation. The Sandia Analysis Workbench (SAW) will be used for workflow control and data management, streamlining the process and bringing discipline and order. In conjunction with DAKOTA, the

SAW will make the UQ process seamless by streamlining building models, submitting jobs, tracking information, managing different versions of codes/data, and managing large amounts of data.

The integrated software development and UB calculation requires parallel workflows of experiments, simulations, and uncertainty propagation across multiple scales. First, the UB workflow within each campaign will provide the necessary input conditions and parameters for the simulation and experiment workflows. If the level of uncertainty is not acceptable, the uncertainty reduction process will be invoked to repeat simulations and experiments. The SAW with DAKOTA workflow management system supports the parallel execution of multiple scientific workflows.

Current workflow managers are focused on running jobs and tracking the output as files. They are not integrated with data management systems that understand the content of those files to provide a querying interface. To support the integration of ML techniques with simulation techniques, we will include a relational database layer for managing the results of running the workflows that tracks both the scheduling of jobs and the data inside the results of the jobs. The workflow manager will produce entity relation diagrams for database schemas, write extract, transform, and load scripts, and support researchers in writing queries for accessing this information to feed ML pipelines based on simulation output.

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